

Arnav Bhilwariya

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SUMMARY

B.Tech IT student at NIT Jalandhar with hands-on experience as a machine learning engineer building end-to-end ML pipelines, data preprocessing systems, and backend APIs. Proficient in Python scripting, scikit-learn, LightGBM, TensorFlow, FastAPI, and cloud-based development workflows.

EDUCATION

National Institute of Technology, Jalandhar 2023 – Present
B.Tech in Information Technology — CGPA: 7.12 / 10

Spring Dales Public School, Muzaffarnagar 2021 – 2022
Senior Secondary (CBSE) – 90%

EXPERIENCE

AD Infocom Systems July 2025 – August 2025
Software Engineer Intern – ML Systems

- Designed **data preprocessing and feature engineering pipelines** using Python scripting, reducing manual overhead by ~35% and cutting ML model training setup time by ~40%
- Developed **3+ FastAPI endpoints** for real-time ML model inference and model deployment; integrated predictions into client-facing REST APIs with Git-based version control
- Engineered automated **data pipelines** to generate model-ready datasets from raw inputs, supporting cross-functional data science workflows and eliminating manual preprocessing steps

PROJECTS

Retail Demand Forecasting | *Machine Learning, Time Series* | [GitHub](#) 2026

- Built **time-series demand forecasting** system predicting sales across 50+ stores and 3,000+ items; trained LightGBM achieving MAE 1.38 and RMSE 2.01, **outperforming baseline by 22%**
- Designed end-to-end **ML data pipeline**: ingestion → feature engineering (lag, rolling, date features) → model training → evaluation using scikit-learn and Pandas
- Applied seasonal decomposition and store-wise demand analysis; visualized prediction intervals to surface **actionable inventory insights** across product categories

Sparrow: AI Voice Coaching Platform | *Machine Learning, Signal Processing* | [GitHub](#) 2025

- Co-developed **AI-driven voice analysis system** extracting acoustic features (pitch, clarity, pacing) via audio signal processing (librosa); delivers real-time structured coaching feedback
- Built **ML classification pipeline** to score speech quality using scikit-learn; Python scripting for feature extraction, model evaluation, and labeled audio data training
- Designed session-level analytics module for **tracking vocal improvement over time**; cross-functional collaboration across audio engineering and backend tracks

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, Java, SQL
ML / AI: Scikit-learn, TensorFlow, Keras, LightGBM, NumPy, Pandas, Matplotlib, Seaborn
Concepts: Machine Learning, Feature Engineering, Time Series Forecasting, Model Deployment, Data Pipeline Design, Python Scripting, Signal Processing, Model Evaluation
Backend & APIs: FastAPI, Flask, Django, Node.js, REST API Design
Tools & Cloud: Git, GitHub, AWS (EC2), Google Colab, Linux (basic)

CERTIFICATIONS

Deloitte Australia – Technology Job Simulation 2025
Forage | Technology consulting, data analysis, forensic technology, and software engineering workflows

AlgoUniversity Graph Theory Programming Camp 2025
Advanced graph algorithms and competitive programming under mentorship of Codeforces Master